

IN4230/3230 Obligatory assignment 1

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A DECLARATION OF AI USAGE

AI has been used for:

1. Clarifying content from plenary exercises and helping me understand how the concepts relate to the implementation.
2. Supplement understanding of course topics, subject-specific terminology and code.
Information was verified with course material, the assignment specification, plenary exercises and third-party sources found through Google search.
3. Providing guidance during debugging and code review, including identifying compilation errors, inconsistencies and possible improvements in code structure.
4. Assisting with structuring the document and reviewing the text for clarity, grammar and spelling errors.

How to run makefile

Build: Make

Clean: make clean

MIP vs IPv4 : How is MIP different from IPv4?

MIP is a minimal network-layer protocol designed for direct communication between neighboring hosts, providing basic addressing and packet delivery over ethernet. IPv4 is a widely used protocol for identifying devices on networks and enabling communication between them. (Geeksforgeeks, 2025)

MIP has the bare minimum needed in the header: Source address, destination address, TTL, SDU length and SDU type. This reduces header overhead allowing for faster processing and potentially lower latency due to it. IPv4 on the other hand contains several more header fields and supports more functionality, including as checksum, identification, fragment offset, protocol information and optional header (NetworkLessons, n.d.). These features make IPv4 more flexible and suitable for general networking, but they also introduce additional header overhead and processing. This does not necessarily mean that one protocol is better than the other, as they serve different purposes and are designed for different environments.

MIP-ARP protocol as a special case

MIP-ARP stands for MIP Address Resolution protocol. Normal MIP communication assumes that the sender knows the destination MAC address, however if that information is missing, the Ethernet frame cannot be sent directly. ARP solves this by broadcasting a request to discover which MAC address is associated with the destination MIP address. Its purpose is therefore to map an MIP address to the MAC address needed to deliver an Ethernet frame on the local network (Fortinet, n.d.). This makes MIP-ARP different from MIP payloads, because it's used to discover link-layer information required before normal communication with another host can take place.

In my implementation, the MIP-ARP logic is handled directly inside the MIPD. This works, but I noticed that it also causes several responsibilities to be concentrated in the same part of the program. A cleaner design could refactor the MIP-ARP functionality into a separate component. The MIPD could ask the component to find the MAC address for a destination MIP address, while the component handles the logic. This would improve separation of responsibilities and make code easier to maintain and understand, although it would also introduce an additional interface between components.

Flow chart for client

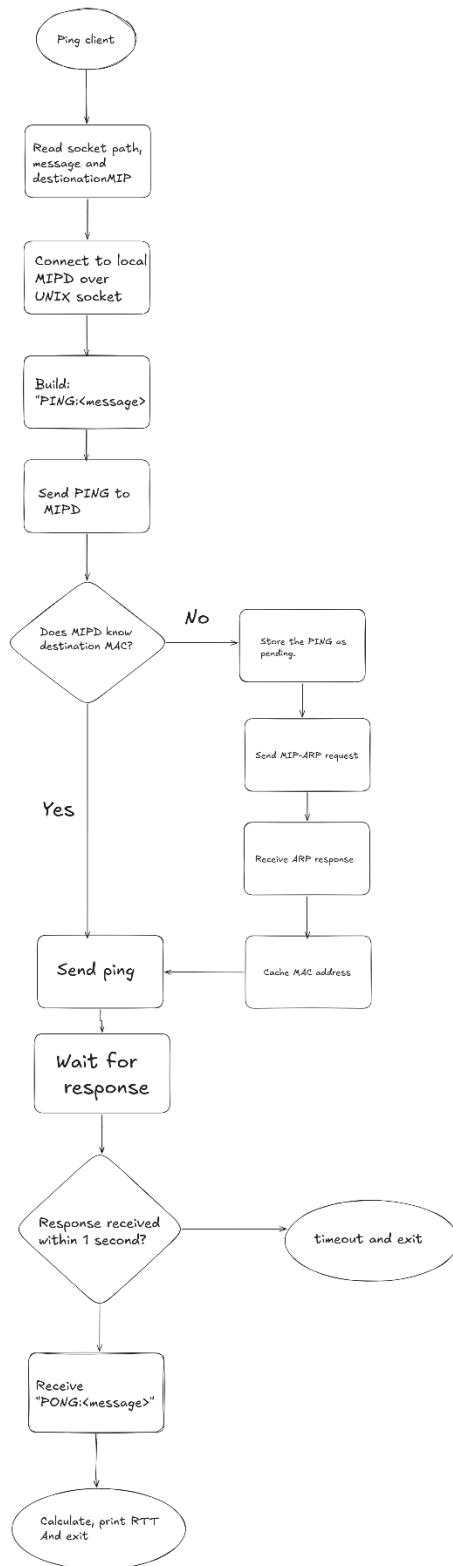


Figure 1: Flow chart of ping_client, made using excalidraw

Flow chart for Server

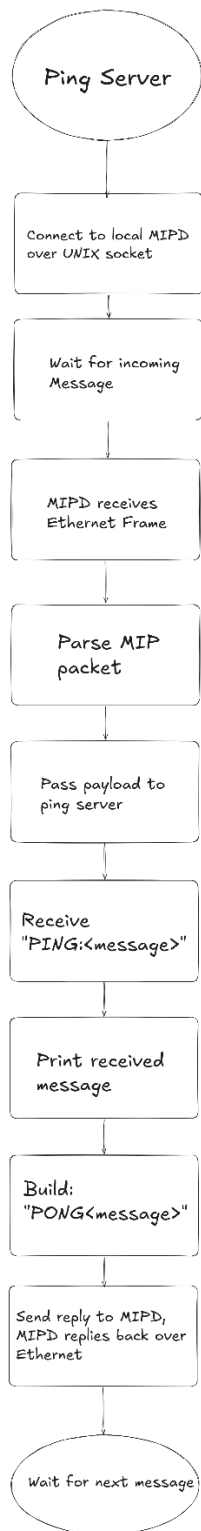


Figure 2: Flowchart of `ping_server` made using excalidraw

References

- Fortinet. (n.d.). *Address Resolution Protocol (ARP): Resolving Network Address Conflicts*. Retrieved from Fortinet: <https://www.fortinet.com/resources/cyberglossary/what-is-arp>
- Geeksforgeeks. (2025, October 4). *What is Internet Protocol Version 4 (Pv4)*. Retrieved from Geeksforgeeks: <https://www.geeksforgeeks.org/computer-networks/what-is-ipv4/>
- NetworkLessons. (n.d.). *IPv4 Packet Header*. Retrieved from Networklessons: <https://networklessons.com/ip-routing/ipv4-packet-header>